

Èric Casanovas Pérez

AI ENGINEER / FULL-STACK SOFTWARE ENGINEER

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SUMMARY

Full-stack engineer working as an AI engineer, 3+ years shipping software in regulated enterprise environments — global payroll, consumer banking and public administration. Sole developer of an LLM-based reconciliation system for a payroll platform operating across 140+ countries, working from architecture through to delivery, as one of the first engineers in a newly created AI team at ADP. Earlier work spans LLM-based document assessment for public-sector recruitment and native mobile development for a consumer bank. MSc in Data Science.

EXPERIENCE

AI Engineer — ADP

Jul 2025 – Present

One of the first engineers in a newly created AI solutions team for ADP Celergo, a global payroll platform operating across 140+ countries.

- **Sole developer for 8 months** (~95% of project commits) of an LLM-based reconciliation system detecting and explaining data discrepancies in multi-country payroll processing, for a process previously handled manually at 3–14 minutes per case plus hours of external correspondence. Covered architecture, API design, data model, prompts, integrations and deployment.
- Engineered reliability into the LLM pipeline: decomposed a single large inference call into scoped subtasks with chunking, plus a validation pass assigning a confidence level to every recommendation — **~90% accuracy with zero false positives** in pre-production validation.
- Designed a shadow-mode evaluation harness to log real operator resolutions as ground truth, so recommendations can be measured against actual human decisions.
- Built structured extraction and entity matching over heterogeneous schemas defined independently by each data provider, across multilingual documents, resolving identity despite typographical errors in source data.
- Prototyped a TF-IDF keyword generator in 2 days for an internal file-validation layer — a deterministic, auditable method chosen over an LLM. Handed over for productisation.
- Write own user stories, scope and priorities with product and business stakeholders, delivering end to end through a heavily governed enterprise release process.

Software Engineer — ADP

Sep 2024 – Jul 2025

- Part of a 15-engineer team owning document management, notification and UI modules used across multiple company applications, including ADP Celergo. Backend APIs and web UI in .NET and TypeScript, inside a compliance-controlled delivery process.

Software Engineer — VASS Consultoría de Sistemas

Sep 2023 – Aug 2024

Consultancy engagement for imagin (CaixaBank Group), a consumer digital bank in Spain.

- Developed native mobile UI components for the imagin app in Java (Android) and Swift / Xcode (iOS), delivering the same components across both platforms.
- Shipped the home-screen card system and a magnetic carousel component, within banking-sector release and compliance constraints.

EXPERIENCE

Software Engineering Intern — Ajuntament de l'Hospitalet

2022 – 2023

- Co-developed an LLM-based document assessment system (GPT-4, via a Microsoft agreement, in early 2023) producing a first-pass provisional score for applicant documentation in public-sector recruitment, which candidates could accept or appeal for human review.
- Designed with a human-in-the-loop escalation path — a use case now classified as high-risk under the EU AI Act.
- Developed backend services in .NET on legacy public administration systems.

EDUCATION

MSc in Data Science — Universitat Oberta de Catalunya

Feb 2024 – Jul 2025

Completed while working full-time. Thesis: a public-grant recommendation system combining automated web scraping, a RAG document store and tool-using agents routing each query to SQL lookups or document retrieval.

BSc in Computer Engineering — Universitat Pompeu Fabra

2019 – 2023

Final project: machine learning for irony detection in text (NLP).

PROJECTS

Write-ups, architecture notes and demos: erico089dev.github.io

ACG — Automatic Content Generator. Content generation and publishing pipeline: 9 microservices with declarative YAML DAG orchestration and health-gated dependencies, a plugin-based flow engine across 8 formats, 4 LLM backends (cloud and self-hosted), 5 TTS engines with voice cloning, CLIP and BLIP-2 with ChromaDB for semantic indexing, serverless publishing via Cloudflare R2 and GitHub Actions.

triagebox. FastAPI service turning voice notes into structured tickets resolved by Claude, with faster-whisper transcription and multi-tenant architecture. Public repository.

Duna — personal assistant. TypeScript monorepo with natural-language calendar, tasks and chat. PWA with Web Push on iOS, SQLite, Tailscale networking. In daily use.

SKILLS

AI & LLMs. LLM orchestration, prompt and context engineering, task decomposition, confidence scoring, shadow-mode evaluation, human-in-the-loop design, tool calling, MCP, LangChain, LangGraph, Azure OpenAI, self-hosted models, Whisper, TTS and voice cloning

ML & Data. RAG, agents and tool routing, TF-IDF, embeddings, semantic search (CLIP, BLIP-2, ChromaDB), NLP, classification metrics

Architecture. Microservices and service decomposition, API design, event-driven and queue-based processing, containerised deployment, CI/CD, technical scoping, spec-driven development, enterprise release governance

Languages. Python, TypeScript, C#, SQL, Java, Swift

Frameworks & infrastructure. FastAPI, .NET, REST APIs, React, Astro, Node.js · Docker, Kubernetes, Jenkins, GitHub Actions, AWS SQS, Azure, Cloudflare R2 · PostgreSQL, Oracle, SQLite

LANGUAGES

Catalan — native · Spanish — native · English — B2, professional working proficiency